

Document_ID

Unique ID: Har document ka `_id` alag hota hai. Same collection me 2 documents ka `_id` same nahi ho sakta.

Type: MongoDB me jo default `_id` banta hai uska type hota hai `ObjectId`.

Automatically Create Hota Hai: Agar aap insert karte time `_id` nahi dete, to MongoDB khud ek `ObjectId` generate kar deta hai.

Primary Key ki Tarah: SQL me jaise primary key hoti hai, MongoDB me `_id` waise hi behave karta hai.

Fast Search: `_id` par automatically index bana hota hai, isliye `_id` se search bahut fast hota hai.

Document Identification: Update, delete, ya specific document find karne ke liye `_id` use hota hai.

Immutable (Usually): Ek baar document create hone ke baad `_id` change nahi kiya ja sakta.

_Id Structure:

In MongoDB, an ObjectId is a 12-byte binary value. The reason you see it as 24 characters is because it is typically represented as a hexadecimal string

- ❑ 1 byte = 8 bits
- ❑ In Hexadecimal, 1 byte is represented by **2 hex characters**

_Id Breakdown:

1. Timestamp (First 4 bytes)

- ❑ Ye batata hai document kab create hua.
- ❑ ObjectId se approximate creation time pata chal sakta hai.
- ❑ Example: agar ObjectId 65f1a2... hai to uska timestamp decode kiya ja sakta hai.

2. Machine Identifier (Next 3 bytes)

- ❑ Ye identify karta hai kaunse machine/server ne document create kiya.
- ❑ Distributed systems me helpful hota hai.

3. Process ID (Next 2 bytes)

- ❑ Ye batata hai kaunsa process document generate kar raha tha.

4. Counter (Last 3 bytes)

- ❑ Ye ek incrementing counter hota hai.
- ❑ Same second me multiple documents insert hone par unko unique banata hai.

**Thank
You**